Collatz Conjecture - A demonstration through the spiral of Theodorus.

Exact construction, Collatz tests and proved return on specified domains

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Structuring the hypotheses, directing the project, and formulating and developing the methodological and theoretical-mathematical framework of this proposal: Davi Cançado Dutra Cota.

Computational execution, test implementation and execution, verification of records, and writing of this version: ChatGPT, under the author’s direction, using one model, no agents and no internet research.

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Abstract

The primary objective is to investigate the Collatz hypothesis using an exact, extensible geometric framework: the spiral of Theodorus. Each positive integer n is represented by a point of quadrance n. The associated family of circles organizes the states, while the Collatz rule determines their transitions. The method extends the test domain while preserving definitions and invariants.

Every start from 1 through 10,000,000 reached 1 in the three recorded implementations. The Theodorus invariant is proved for every index, and return is proved for an explicitly parameterized infinite family. The hypothesis is confirmed on the examined domain; convergence for all positive integers still requires proof.

A complementary geometric study proves a tangent spatial passage whose projection reaches the apparent center. It examines depth and the adequacy of representation through its own proof. The author interprets the compatibility of these results in philosophical dialogue with Socrates and Matthew 19:23-26.

Keywords: Collatz; Theodorus; quadrance; exact arithmetic; verification; projection; philosophy of mathematics.

1. Objective: investigate Collatz through Theodorus

Davi Cançado Dutra Cota selects Theodorus as a methodological framework for testing the Collatz hypothesis. The primary question is whether every positive-integer trajectory generated by the following rule reaches 1 after finitely many steps. Constructing the spiral and updating a Collatz state are separately defined operations.

$$C(2k) = k, \qquad C(2k+1) = 6k + 4$$

$$\forall n \in \mathbb{Z}_{>0} \quad \exists j \in \mathbb{Z}_{\geq 0} : \quad C^j(n) = 1$$

In the first formula, k≥1 in the even branch and k≥0 in the odd branch. Reaching 1 means reaching integer state 1; continuing the rule produces the loop 1→4→2→1. In Theodorus, that state is $v_1=(1,0)$, of quadrance 1. The origin has quadrance zero.

Why select an extensible construction

Each construction step adds a unit leg perpendicular to the preceding radius. The identity Q(v_N)=N associates every index with a circle $x^2+y^2=N$ and an exact point on it. The rule extends to any finite index; perfect-square indices have integer radii. This organization supports larger tests without changing what the states mean.

Part of the investigation	Question and intended result
Theodorus and Collatz	Represent states and check whether trajectories of rule C reach 1.
Complementary spatial geometry	Prove a tangent passage and identify the depth omitted by its projection.
Philosophical interpretation	Relate possibility, knowledge and choice to results of explicitly stated scope.

How the article uses the word demonstration

The text proves the general invariant, return for an infinite family and the spatial construction; it records confirmation of Collatz on an exhaustive finite interval. Universal convergence remains an additional proof requirement. The purpose of using Theodorus is to investigate Collatz. The spatial passage belongs to the complementary analysis of representations and has its own geometric and trigonometric proof.

Methodological robustness here means stable definitions, exact certificate arithmetic, checkable invariants, controls and reproducible records. The article presents this method and its results first, followed by spatial depth, applications and philosophical interpretation.

2. A generating pattern with no index ceiling

The value of choosing Theodorus starts with its construction. Beginning at $v_1=(1,0)$, add $u_N=(-y_N,x_N)/\sqrt{N}$ to $v_N=(x_N,y_N)$. By induction, u_N is unit and perpendicular to v_N, with consistent orientation. Each new triangle repeats the local rule although its central angle changes. Define $Q(x,y)=x^2+y^2$.

$$v_{N+1} = v_N + u_N, \quad v_N \cdot u_N = 0, \quad Q(u_N) = 1$$

$$Q(v_{N+1}) = Q(v_N) + 1 \quad \Longrightarrow \quad Q(v_N) = N \quad (N \geq 1)$$

The second identity holds at every index by induction: it holds at N=1, and the rule preserves the unit increment. This gives a precise meaning to infinite expansion: the formula extends to any finite index without rounding quadrance. Infinitely many executed steps are unnecessary to prove this invariant.

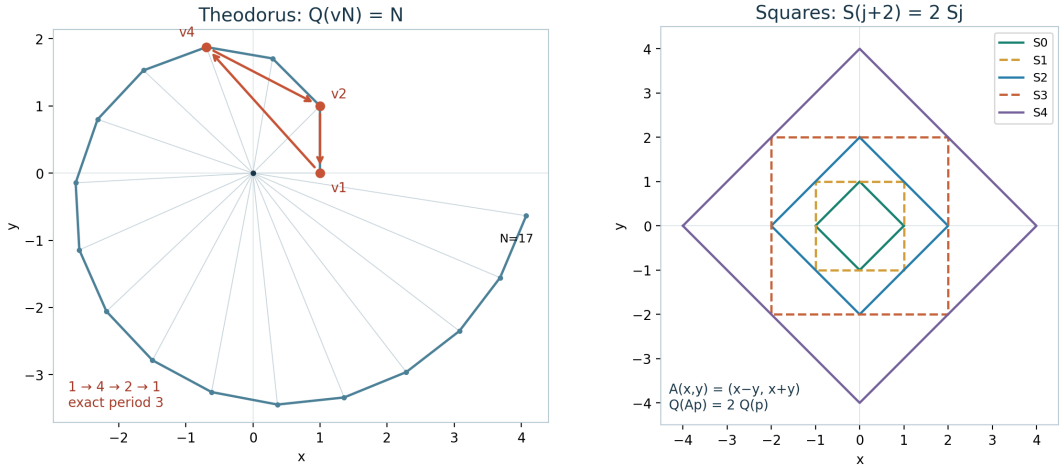


Figure 1. Left: Theodorus and the encoded Collatz loop. Right: an exactly self-similar square family, distinct from the entire spiral. [R12]

Three properties with precise names

The construction has a geometric recurrence with exact invariants. The loop 1→4→2→1 has exact periodicity of period 3. The square family $S_j=A^jS_0$, where $A(x,y)=(x-y,x+y)$, has exact self-similarity: $Q(Ap)=2Q(p)$ and $S_{j+2}=2S_j$. Here S_0 has vertices $(\pm1,0),(0,\pm1)$. Expanding the two squares proves the first identity; $A^2(x,y)=2(-y,x)$, together with quarter-turn invariance of each square, proves the second. Both hold at every level.

The angle between consecutive Theodorus rays has squared sine $1/(N+1)$, so it changes with N. Square self-similarity does not mean that the whole spiral is a scaled copy at every step, or that every four triangles complete a turn.

3. Why Theodorus is a valuable testing framework

Theodorus is selected to structure Collatz testing: one state for each integer, a common generating rule and checkable local invariants. Robustness in this study rests on preserving the rule, using exact certificate arithmetic and reproducing the records. These criteria make the construction’s methodological role explicit.

Resource	Role in the method
Identifier Q(v_N)=N	Recovers the represented integer exactly. Distinct quadrances guarantee distinct states.
Orthogonality and unit step	Check each extension by dot products and norms instead of inferring exactness from a picture.
The same rule at every index	Expand the domain without changing the definition, keeping results comparable across scales.
Powers and squares	Organize state families and halving descents, suggesting hypotheses that require proof.
Other rules and curves	Separate properties of the drawing from properties of 3n+1. Archimedes provides a control here.

A test must preserve the studied rule

$$E(n) = v_n, \qquad F(E(n)) = E(C(n))$$

Since E is injective, a trajectory reaches E(1) exactly when its integer reaches 1. Repetition is preserved too. This makes the representation auditable: an implementation that loses a loop or invents an integer exit fails the correspondence test. The construction step N→N+1 and the dynamical jump n→C(n) are different operations.

Infinite extension and finite experiments complement each other

Induction certifies Q(v_N)=N for every N; enumeration certifies the trajectories actually covered. Tests can grow without rounding integer states. Integer size, memory and time still grow, however. Full radical coordinates were not computed for ten million spiral vertices: the test retains their exact identifiers.

The contribution is an infrastructure for investigation: formulate a hypothesis, declare the encoding, check invariants and gates, execute a specified domain and preserve reproducible records. Universal convergence still requires an argument for the destination of every trajectory, beyond the ability to represent it.

4. An infinite family with proved return

Theorem and domain

For every integer k≥2 and a≥0, the starting value n=$2^a(4^k-1)/3$ reaches 1 under the usual Collatz rule. Count one step for each halving and one for each odd-branch application, stopping at the first arrival at 1.

$$p_k = \frac{4^k - 1}{3}, \qquad n = 2^a p_k, \qquad \tau(n) = a + 1 + 2k$$

Proof

Because 4 is congruent to 1 modulo 3, p_k is an integer. Its numerator and denominator are odd, so p_k is odd. The first a halvings remove exactly the factor 2^a. Since k≥2, p_k≥5 and the trajectory has not reached 1. The next odd step gives 3p_k+1=$4^k=2^{2k}$. Exactly 2k halvings then reach 1, with no earlier value equal to 1 on that descent. This proves the first-arrival formula.

$$2^a p_k \longrightarrow p_k \longrightarrow 4^k \longrightarrow 2^{2k-1} \longrightarrow \cdots \longrightarrow 1$$

k	a	Starting value n	Steps to 1
2	0	5	5
2	3	40	8
3	0	21	7
7	4	87,376	19
1,025	255	$2^{255}(4^{1025}-1)/3$	2,306

Theodorus and the square interpretation

Every state has a vertex v_n of quadrance n. The gate 4^k has radius 2^k, an exact integer. Each subsequent halving halves quadrance. The square family represents this proportion without rounding. The arithmetic proof and its geometric interpretation certify the same sequence.

The proof covers infinitely many starts, beyond any executed interval. It is an explicitly parameterized family, not every positive integer. Proving that every start reaches such a descent remains the additional requirement. No new priority is claimed for this elementary consequence of the gate form.

The execution checked 262,144 block certificates for 2≤k≤1,025 and 0≤a≤255, and retained 60 full trajectories from a declared sample. Exact block certification and step-by-step enumeration are different procedures and are recorded accordingly. [R13]

5. Executed expansion and a checkable certificate

The protocol was recorded before execution. Every start from 1 through 10,000,000 was checked using three integer implementations: the direct rule, compressed halvings with exact step counts, and square state $q=n^2$ with exact integer roots. The methods share the certificate infrastructure disclosed in the code. [R13]

Measurement	Result
Distinct starts	10,000,000; all reached 1
Disagreements / unresolved / overflow	0 / 0 / 0
Maximum steps to 1	685; first start attaining it: 8,400,511
Largest visited integer	60,342,610,919,632
Certificates from the three methods	Identical byte for byte
Additional Python verifier	10,000,000 recurrence relations and gates checked
Altered certificate controls	5 deliberate errors; all 5 rejected

How the certificate supports the finite conclusion

Each record contains the first-arrival time and exponent of the first gate 4^k; starts already equal to a power of 2 carry marker 0. A separate verifier recomputes excursions outside the interval and checks $t(n)=\text{steps}+t(\text{reentry})$, with $t(1)=0$ and $t(n)>0$ for $n>1$. The strictly decreasing count certifies arrival at 1 throughout the declared domain. The verifier performed 23,625,961 transition evaluations in these excursions.

Expanding exact proportion

The family $S_j=A^jS_0$ was checked at 4,097 consecutive levels $j=0$ through 4,096: vertices, sides, areas, contacts and two-level doubling. Three additional isolated tests checked $j=10{,}000$, 100,000 and 1,000,000 by integer matrix exponentiation. The largest quadrance was exactly $2^{1000000}$, stored as a binary integer with 1,000,001 bits.

The 262,144 family certificates from Section 4 were also checked. Records were reopened and verified; protocols, formats, summaries and code are included. Ten million is an exhaustive finite interval. The three extreme indices do not constitute one million consecutive levels. This expansion strengthens auditability and empirical coverage; it does not replace the still-needed universal proof.

6. Theodorus, positivity and neighbouring rules

Proposition: positive starting values cannot reach zero

Consider the three rules in the manuscript: divide n by 2 when n is even; when it is odd, apply respectively 3n+1, 3n−1 or 5n+1. If n is positive and even, n≥2 and n/2≥1. If n is positive and odd, n≥1, giving the following lower bounds for the odd branch.

Odd-branch rule	Lower output bound	One verified loop
3n+1	4	1→4→2→1
3n−1	2	5→14→7→20→10→5
5n+1	6	13→66→33→166→83→416→208→104→52→26→13

Each rule therefore preserves positive integers. Induction on the number of steps proves that no trajectory starting at n≥1 reaches zero, regardless of its duration or whether it enters a loop.

$$n_0 \in \mathbb{Z}_{\geq 1} \quad \Longrightarrow \quad n_j \geq 1 \quad (j = 0, 1, \ldots)$$

What the Theodorus representation establishes

The identity Q(v_N)=N represents each positive state exactly. It also represents the neighbouring rules; it does not, by itself, select a unique dynamics. The spread 1/(N+1) is positive for each finite N. Its zero limit as N grows does not mean that the integer N has reached zero.

The gate theorem describes the first power of 2 reached by a trajectory that reaches one: 4^k in 3n+1, $2 \cdot 4^k$ in 3n−1 and 16^k in 5n+1, under the manuscript's assumptions. These are distinct arithmetic signatures. Square gates also exist in 5n+1 alongside its loops, so this property alone does not establish universal convergence or uniqueness of Collatz. [T14]

Local checks recorded for this revision

The check covered 30,000 transitions: one for each n from 1 to 10,000 under each of the three rules. Each output was calculated in three exact representations, giving 90,000 evaluations with no disagreements or nonpositive outputs. Ten selected trajectories also agreed across the representations. This finite check verifies the implementation; it does not enumerate every trajectory. [R11]

7. Archimedes as a controlled comparison

The Archimedean spiral relates radius linearly to angle: r=a+bθ. This comparison explicitly chooses a=0 and b=1, with θ≥0 in radians. Two samplings are examined: θ=n, giving Q=n², and θ=√n, giving Q=n. The second matches Theodorus quadrance at each integer.

$$W(\theta) = (\theta\cos\theta, \theta\sin\theta), \quad Q(W(\theta)) = \theta^2$$

$$E_L(n) = W(n), \qquad E_Q(n) = W(\sqrt{n}), \qquad Q(E_Q(n)) = n$$

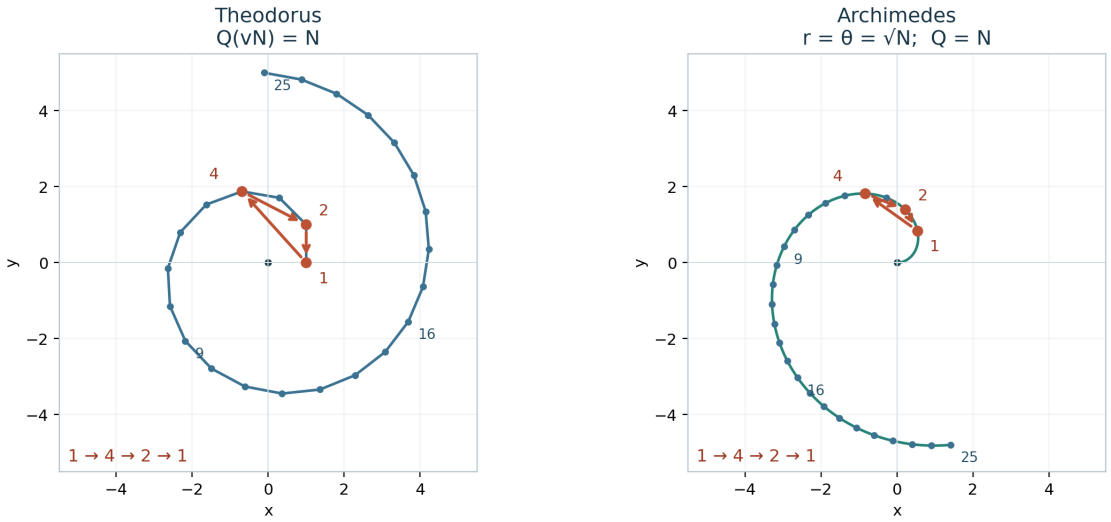


Figure 2. The same indices 1 through 25 and the same quadrance n. Positions change; the loop 1→4→2→1 remains in both encodings. Arrows show transitions, not mandatory routes along the curve.

What the comparison can test

Theodorus combines Q=n, orthogonality and a unit step in its construction. Archimedes has a constant radial increment per constant angular increment. Resampling at θ=√n also gives Q=n, but does not preserve the perpendicular unit step. Thus Q=n is available on more than one curve; the combination of local invariants distinguishes Theodorus.

States and inequalities were handled symbolically and with exact rational bounds, without a numerical value of π. Sine and cosine appear in the definition and drawing. Screen coordinates are declared approximations and do not decide equality, tangency or return. [R14]

8. Archimedes: results and interpretation

Recorded test [R14]	Result
1,000,000 transitions, three encodings	Identical integer successors; zero disagreements
1,000 complete trajectories, starts 1-1,000	59,542 transitions per encoding; all reached 1
10,000 consecutive pairs sampled at θ=n	Every step has quadrance greater than 1; none is perpendicular to its initial radius
10,000 consecutive pairs sampled at θ=√n	Every step has quadrance between 0 and 1 and positive radial dot product
4 scales; 1,000 indices per scale	4,000 cases; exact quadrances preserved after decoding
5 deliberately invalid controls	All 5 rejected

Angular certification by rational bounds

For θ=n, the angular increment is 1: 1/2<cos(1)<13/24. The squared step is therefore 1+2n(n+1)(1−cos(1))>1. Its dot product with the initial radius is positive at n=1 and negative for n≥2, never zero.

For θ=√n, let s=√(n(n+1)) and δ=√(n+1)−√n. The dot product D of the initial radius and the step satisfies the inequality below, using cos(δ)≥1−δ²/2. Positivity follows by comparing squares of the two positive terms: n⁴−(n−1/2)²n(n+1)=n(3n−1)/4>0.

$$D = s\cos\delta - n \;\geq\; n^2 - (n-\tfrac{1}{2})s > 0, \qquad Q(\Delta E_Q) = 1 - 2D < 1$$

The step is also strictly positive because the radii differ. Tests certified these signs using square-root intervals from integer roots and rational cosine bounds, without rounding decisions. The proved inequalities hold for every n≥1; the execution covers the declared 10,000 pairs.

A nonperiodic spiral can represent an exact loop

Both Archimedean encodings are injective because radius identifies n. If F(E(n))=E(C(n)), then F³(E(1))=E(1), even though the supporting curve has increasing radius and is not periodic. Nonperiodicity alone does not create a new Collatz transition. This comparison supports distinguishing construction, encoding and rule, and makes the methodological value of Theodorus more specific.

9. Projection and depth: complementary study

Following the investigation of Collatz through Theodorus, this complementary study examines the adequacy of spatial representations. Davi Cançado Dutra Cota asks what becomes invisible when a spatial structure is shown in a planar view. A tetrahedron provides an example: its edges can appear as crossing diagonals while occupying different heights. [D1]

$$P(x, y, z) = (x, y), \qquad P^{-1}(0, 0) = \{(0, 0, z) : z \in \mathbb{R}\}$$

The apparent center therefore does not identify a unique spatial point. It can represent the sphere center, an octahedron pole or an exterior point on the z-axis. A passage through the projection fiber over the center makes precise the intuition of an exit hidden by the image.

Object	Definition in this article
Equatorial circle Cρ	z = 0 and x² + y² = ρ. The planar boundary.
Disk Dρ	x² + y² ≤ ρ in the xy-plane. Includes its interior.
Sphere Sρ	Q(x,y,z) = x² + y² + z² = ρ. The surface.
Ball Bρ	Q(x,y,z) ≤ ρ. The solid used in the collision test.

The projection of the entire spherical surface is the disk; its silhouette is the circle. The latter is a closed set: if its points converge in the plane, continuity of x²+y² preserves equality with ρ at the limit. Depth information remains absent from the image.

A complete circular image can come from an open spatial path

$$q(u) = \left(\frac{1-u^2}{1+u^2}, \frac{2u}{1+u^2}\right), \quad 0 \le u \le 1, \quad J(x, y) = (-y, x)$$

The identity $(1-u^2)^2+4u^2=(1+u^2)^2$ proves $Q(q)=1$. For $j=0,1,2,3$, the path $\gamma(j+u)=(r\,J^j q(u), h(j+u)/4)$, with $r,h>0$, traverses all four quarters of the circular image while gaining height. Its endpoints are $(r,0,0)$ and $(r,0,h)$. Joins are continuous because $q(1)=Jq(0)$. The image completes a turn while the spatial path has distinct endpoints. This construction proves the information loss examined in the complementary spatial study.

10. Construction of the tangent spatial passage

This part examines what projection preserves and what it omits. The spatial proof has its own coordinates and conditions; its role in the project is to make depth and representational adequacy explicit. Collatz testing was developed through Theodorus in the preceding sections.

$$\rho = \frac{7}{2}, \quad T = (1, \frac{3}{2}, \frac{1}{2}), \quad X = (0, 0, 7)$$

$$d = X - T = (-1, -\frac{3}{2}, \frac{13}{2}), \quad B(t) = T + td, \quad 0 \le t \le 1$$

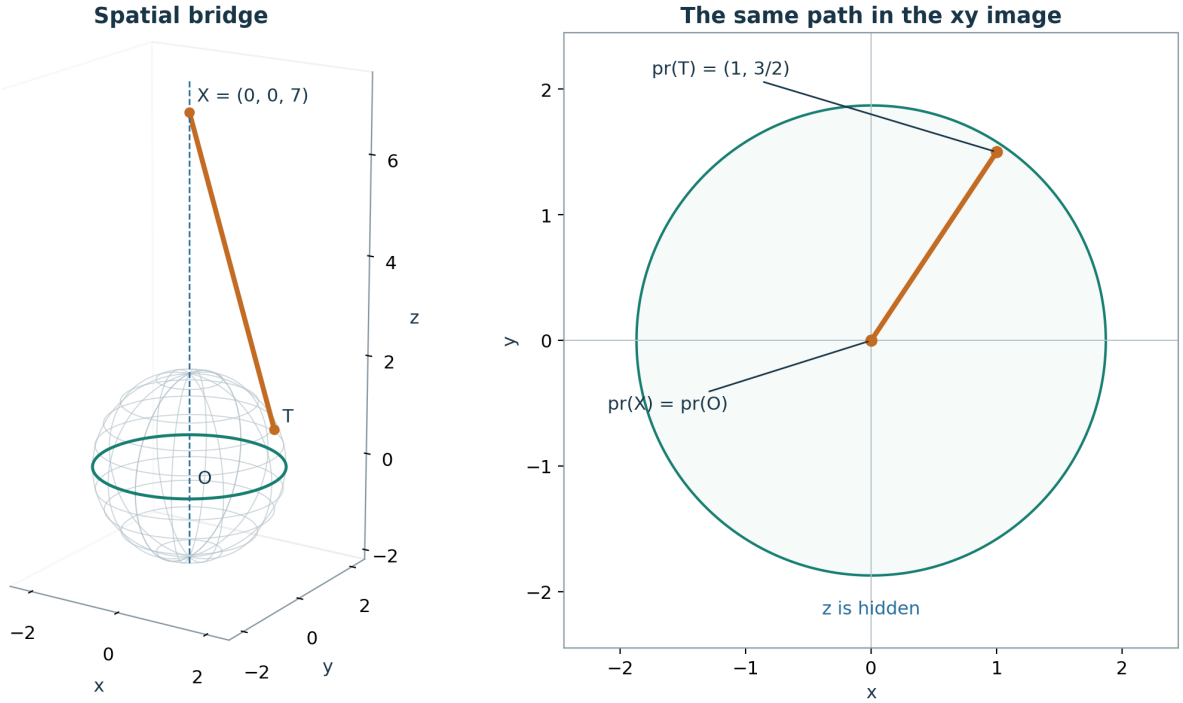


Figure 3. Two views of the bridge T→X. The point X lies outside the sphere and projects to the center. The green circle lies in z = 0. These are rasterized illustrations; the coordinates and identities provide the proof.

The sphere has radius √14/2 and diameter √14. The point T lies on the surface because 1 + 9/4 + 1/4 = 7/2. The point X has quadrance 49 and is exterior. The segment's image runs from (1, 3/2) to (0, 0), hiding the increase in z.

This is an explicit realization of the bridge idea with checkable assumptions. Passage through the center means reaching the axis projected to the center; the segment does not pass through the spatial center O. Tangency will be established with respect to the sphere.

11. Tangency, inversion and the contact cone

$$Q(T) = \rho, \qquad T \cdot d = 0, \qquad Q(d) = \frac{91}{2}$$

The normal to the sphere at T is parallel to T. Since T·d = 0, the line in direction d is tangent at T. Furthermore, Q(T+td) = ρ + t²Q(d): the line meets the surface only at T and is exterior at every other point.

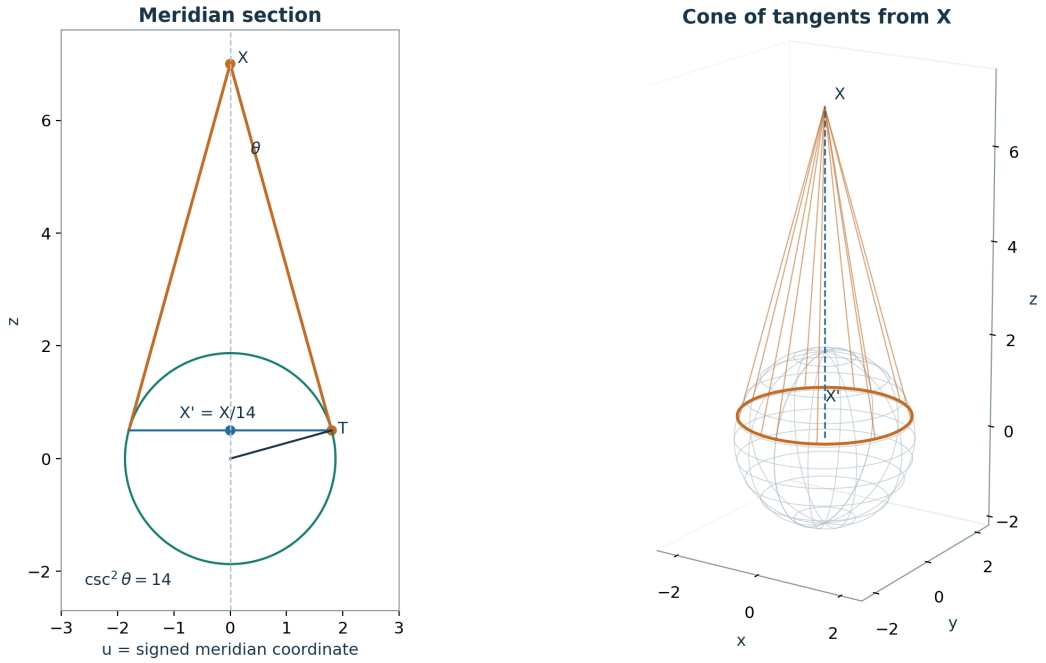


Figure 4. Left: a section through the z-axis and a tangent; u is a signed meridian coordinate. Right: the cone of all tangents from X to this sphere.

$$X' = \frac{\rho}{Q(X)} X = \frac{X}{14} = (0, 0, \frac{1}{2})$$

For any contact point U, the condition U·(X−U) = 0 gives U·X = ρ. Thus z = 1/2 and x² + y² = 13/4. The center of this contact circle is the inverse point X′. Every tangent in this cone has the same angular ratio. This precisely relates a family of tangents; it is not a claim about every tangent in every configuration.

Contact T lies above the equatorial plane. Its image is inside the disk: x²+y² = 13/4 < 7/2. The bridge line is tangent to the sphere, while its projection is not tangent to the equatorial circle.

12. The angular role of √14 and the axial phase

For nonzero vectors, an angular ratio can be recorded without evaluating trigonometric functions. Define s(d,e) = 1 − (d·e)²/[Q(d)Q(e)]. This is the squared sine of the angle between the directions; the sign of the dot product distinguishes acute from obtuse orientation.

$$s(d, e_z) = \frac{1 + 9/4}{91/2} = \frac{1}{14}, \quad \csc\theta = \sqrt{14}$$

$$\sin\theta = \frac{1}{\sqrt{14}}, \quad \cos^2\theta = \frac{13}{14}, \quad d \cdot e_z > 0$$

Phase	Reference direction	Sine and cosecant
Bridge T→X	d relative to +ez	sine = 1/√14; cosecant = √14
Continuation X+s ez	+ez relative to +ez	sine = 0; cosecant undefined
Instant of rest	Zero velocity	The velocity angle is undefined

In right triangle OTX, the cosecant also equals OX/OT = 7/(√14/2) = √14. One can continue from X along z with zero sine relative to the axis, in a second phase. Zero sine and a finite cosecant are not assigned to the same angle.

Continuous position, velocity and acceleration at the join

$$f(t) = 10t^3 - 15t^4 + 6t^5, \quad f'(t) = 30t^2(1-t)^2$$

$$f''(t) = 60t(1-t)(1-2t)$$

Use T+f(t)d in the first phase and X+h f(u)ez in the second, with 0≤t,u≤1 and h>0. Velocity and acceleration vanish at both ends of each phase, allowing a C² join with a stop at X. Rational parameters give rational points. This proves regularity of the idealized motion; friction, actuator limits and measurement precision require their own models.

13. The tetrahedron: square image, distinct heights

$$c = \frac{7}{\sqrt{8}}, \quad A = (c, c, c),\ B = (c, -c, -c),\ C = (-c, c, -c),\ D = (-c, -c, c)$$

Each vertex pair differs in two coordinates by magnitude 2c. All six edges therefore have length √(8c²) = 7: the tetrahedron is regular. Its orthogonal xy projection has vertices (±c,±c), forming a perfect square.

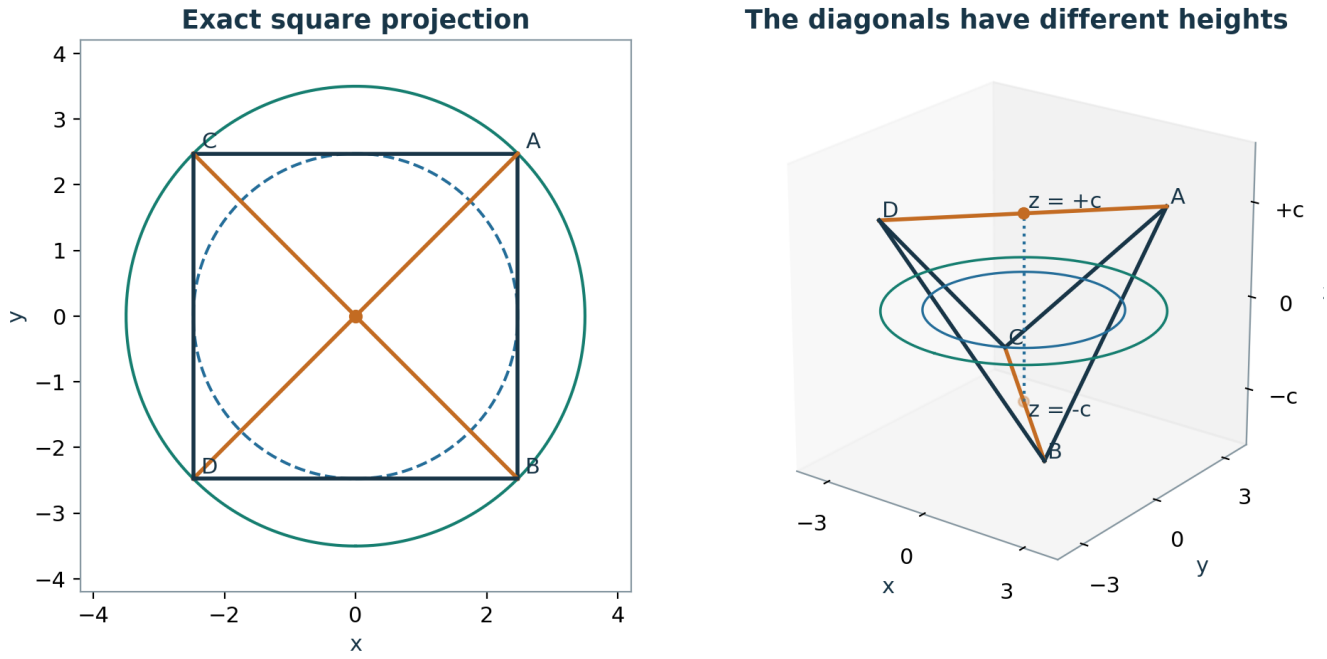


Figure 5. Edges AD and BC appear as internal diagonals. Their midpoints are (0,0,+c) and (0,0,−c), separated by distance 2c. Coincidence in projection is not a spatial intersection.

The circle circumscribed about the image has radius √2c = 7/2 and passes through its four projected vertices. The inscribed circle has radius c, and the projected square sides are tangent to it. These are image properties. The spatial vertices have z = ±c and do not lie on the circle drawn in z = 0.

This demonstration has its own purpose: to show directly how an apparent intersection can conceal depth separation. It does not identify a planar square with a unique tetrahedron; different spatial objects can produce the same image. Nor does it independently define a transition of Collatz integers.

14. The octahedron, squares and the √14 scale

$$R = \frac{\sqrt{14}}{2}, \quad a = \frac{\sqrt{7}}{2}, \quad R^2 = 2a^2 = \frac{7}{2}$$

Take four equatorial vertices (±a,±a,0) and poles (0,0,±R). All twelve octahedron edges have quadrance 7. The equatorial square diagonal and the pole-to-pole distance are both √14. The sphere containing all six vertices is the one used for the bridge.

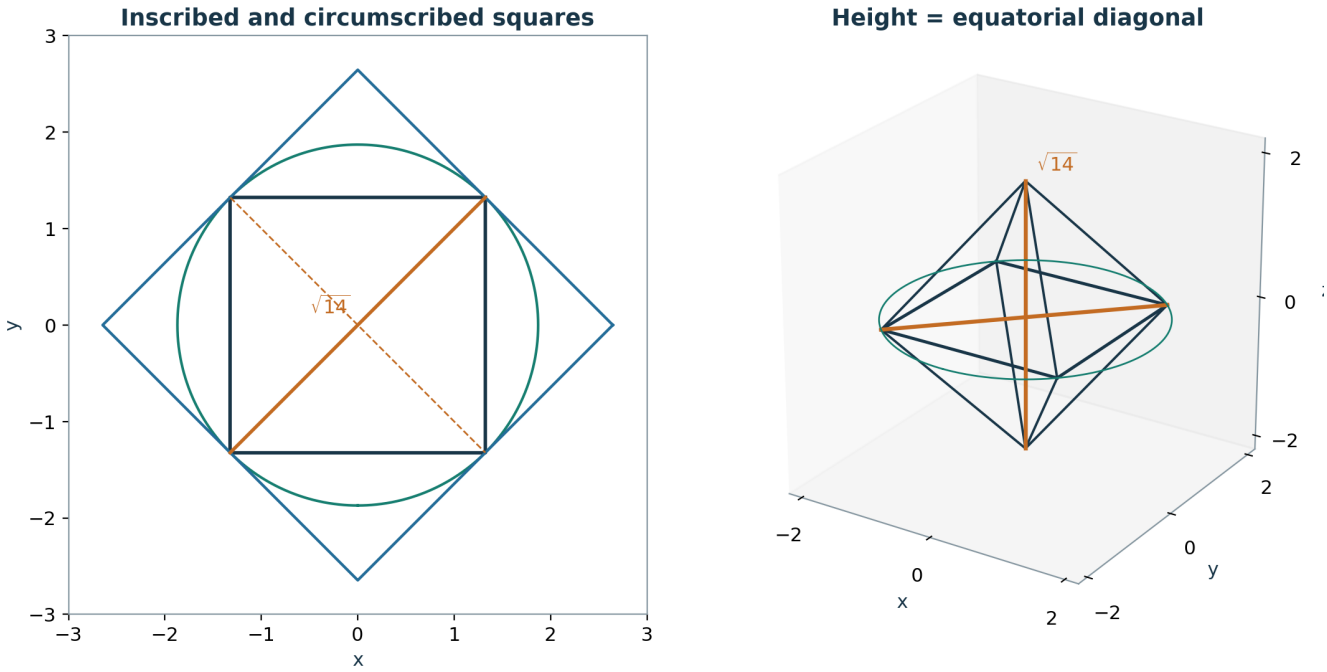


Figure 6. The inner square has side √7 and diagonal √14. The outer square, rotated 45° relative to the inner one, has side √14 and is tangent to the circle at the inner square's four vertices. Right: the octahedron's total height is √14.

The outer square has vertices (±√7,0) and (0,±√7). Its sides lie on ±x±y = √7; their distance from the origin is √(7/2). This proves circumscription using dot products and quadrances.

The perpendicular squares sharing a diagonal, in the Cota definition supplied by the proponent, can be realized in sections of a regular octahedron. [D1] The common sphere associates this figure with the bridge. The bridge contact T was not chosen as an octahedron vertex, and no such identification is required by the proof.

15. An exact difference between section and surface

Proposition: the equatorial section has no rational point

For the selected radius, $x^2+y^2 = 7/2$ has no solution with both x and y rational. However, the sphere $x^2+y^2+z^2 = 7/2$ has the rational point T = (1,3/2,1/2). This demonstrates a genuine arithmetic consequence of adding a coordinate.

$$x = \frac{a}{c}, \quad y = \frac{b}{c} \quad \Longrightarrow \quad 2(a^2 + b^2) = 7c^2$$

Proof. Choose integers a,b,c with no common factor and $c \neq 0$. The quadratic residues modulo 7 are 0,1,2,4. Two such residues sum to zero modulo 7 only when both are zero. The equation therefore implies that 7 divides a and b. Substituting a=7u and b=7v gives $c^2=14(u^2+v^2)$, so 7 also divides c, a contradiction. The rational point on the sphere is verified by 1+9/4+1/4=7/2.

Scale and angular ratio

$$T_\lambda = \lambda T, \quad X_\lambda = \lambda X, \quad \rho_\lambda = \lambda^2 \rho, \quad s = \frac{1}{14}$$

Every $\lambda > 0$ preserves tangency and angle. Rational λ keeps all bridge data rational. The ratio $\sqrt{14}$ is independent of the length unit. Its role here is a specific proportion between axis, radius and tangent, not a property exclusive to one absolute sphere size.

This does not imply that every spatial geometry has rational coordinates. The octahedron presented here uses exact radicals. An irrational number such as $\sqrt{14}$ can be represented symbolically without error; irrationality and inexactness are distinct concepts. Quadrances avoid extracting roots when a decision depends only on sums, products and comparisons.

16. The passage exists; taking it is a separate condition

The proponent clarifies that his geometric thesis asserts the existence of a spatial passage, without requiring every trajectory to take it. Depth is hidden by the xy projection. In the proved construction, B ends at X=(0,0,7), outside the sphere, while P(X)=(0,0).

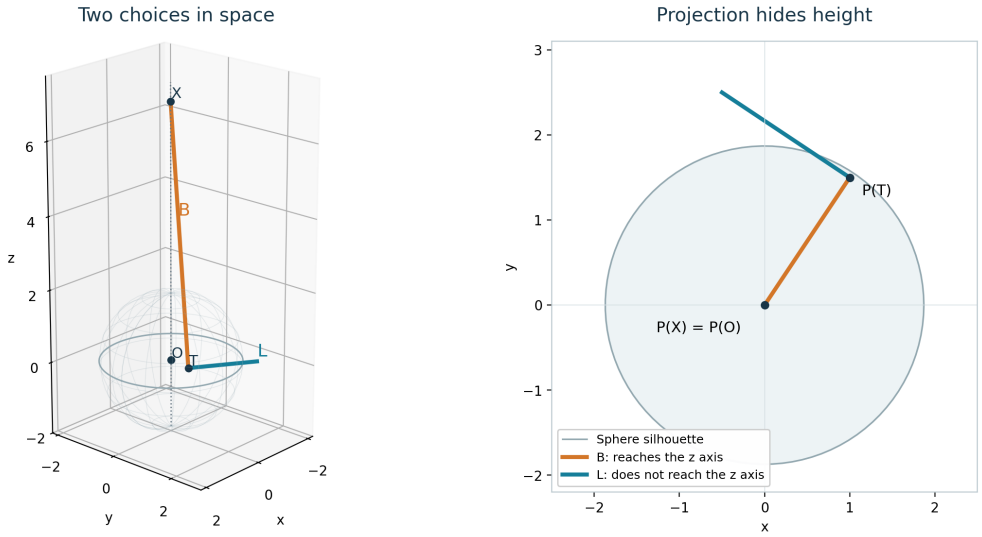


Figure 7. Two paths tangent to the same sphere at T. B reaches the z axis; L takes another direction. Projection places O and X at the same apparent center.

$$u = (-\frac{3}{2}, 1, 0), \quad L(t) = T + tu, \quad 0 \le t \le 1$$

$$T \cdot u = 0, \quad Q(L(t)) - \frac{7}{2} = \frac{13}{4}t^2, \quad Q(P(L(t))) = \frac{13}{4}(1 + t^2) > 0$$

Thus L also leaves tangentially into the exterior but never reaches the z axis. These identities hold throughout the stated real interval. B and L show that existence of the passage does not force its use among these admissible paths. The local check recorded 2,002 rational states of the two paths. [R11]

Relationship to the Collatz methodology

Paths B and L make it possible to check the distinction between spatial information and its planar image. Arithmetic gate 4^k and spatial passage B have their own definitions. Integer return and geometric existence can both hold without a Collatz trajectory having to select B.

The passage is demonstrated geometrically and trigonometrically through exact identities. Its existence complements the analysis of representations. The Theodorus experiment preserves rule C and investigates arrival at state 1; that is its methodological objective.

17. Application: collision decisions that retain depth

Suppose a tool must decide whether a point lies inside the ball Bρ. The correct decision in this model depends on the sign of Q(P)−ρ. If software retains only the xy view and treats the projected disk as the solid, it loses information needed to decide some cases.

$$Q(B(t)) - \rho = \frac{91}{2}t^2$$

$$Q(P(B(t))) = \frac{13}{4}(1-t)^2 < \frac{7}{2} \quad (0 \le t \le 1)$$

At every t>0 along the bridge, the point lies outside the sphere while its projection stays inside the disk. At t=0 there is contact. The proof covers the declared real interval; the test grid checks an implementation on a finite collection of rational parameters.

Why xy alone cannot decide every case

The points O=(0,0,0) and X=(0,0,7) have the same projection. The first lies inside the ball and the second outside. A classifier receiving only (0,0) gives the same answer for both, and therefore cannot classify both correctly. The limitation comes from losing a relevant variable, independently of algorithm speed.

This observation is useful for visualization, trajectory planning and interference checking. The planar test can serve as a preliminary filter followed by a spatial decision. If the obstacle is an infinite cylinder parallel to z, an xy test may suffice. The result depends on the object whose collision is being assessed.

What is certified

The prototype certifies the sign of a rational expression for rational input. This exactness covers the specified geometric decision. Unit errors, uncertain measured coordinates, an inappropriate obstacle model and defects outside the checked kernel require separate treatment.

18. Protocol, samples and records

The protocol was recorded before execution, including hashes of both calculation scripts. All tests ran locally, without networking, auxiliary agents or calls to other models. Scales are $\lambda=2^{e}$ for all 33 integers e from −16 to 16. [E1]

Experiment	Declared domain	Result
Bridge and projection	33 scales × 1,001 values t=j/1,000	33,033 states: 33 contacts and 33,000 exterior points
Tangency stress	33 scales × $t=2^{-k}$, k=1…600	19,800 exact positive signs
Radial controls	$(1\pm2^{-k})T$, k=1…600	1,200 cases on both sides of the surface
Input contract	33 valid proposals + 12 changes at each scale	33 accepted; 396 invalid proposals rejected
Smooth transition	t=j/1,000, j=0…1,000	1,001 samples; endpoints and derivatives checked
Calculation time	12,000 inputs; 3 alternating-order passes	Identical exact outputs in all passes

The grid was fully enumerated: “exhaustive” refers to this explicitly bounded finite domain. These tests do not cover all programs, all numbers or all AI models. Analytical proofs establish the general identities; execution records the code’s agreement in the tested cases.

Compressed files retain one row per case. An integrity check found one first-run file truncated. It was preserved as history and the unchanged code was rerun. Complete repeated-run records were checked and adopted for this edition; mathematical counts agreed. The incident is documented in registros/intercorrencia_integridade.json.

19. What the error tests actually measure

Stress-test method	Incorrect signs / 19,800	Occurrence
Coordinates in binary64	18,843	18,810 zeros and 33 negative signs
Invariant in binary64	1,980	1,980 zeros from underflow
Exact fractions	0	The positive sign was preserved

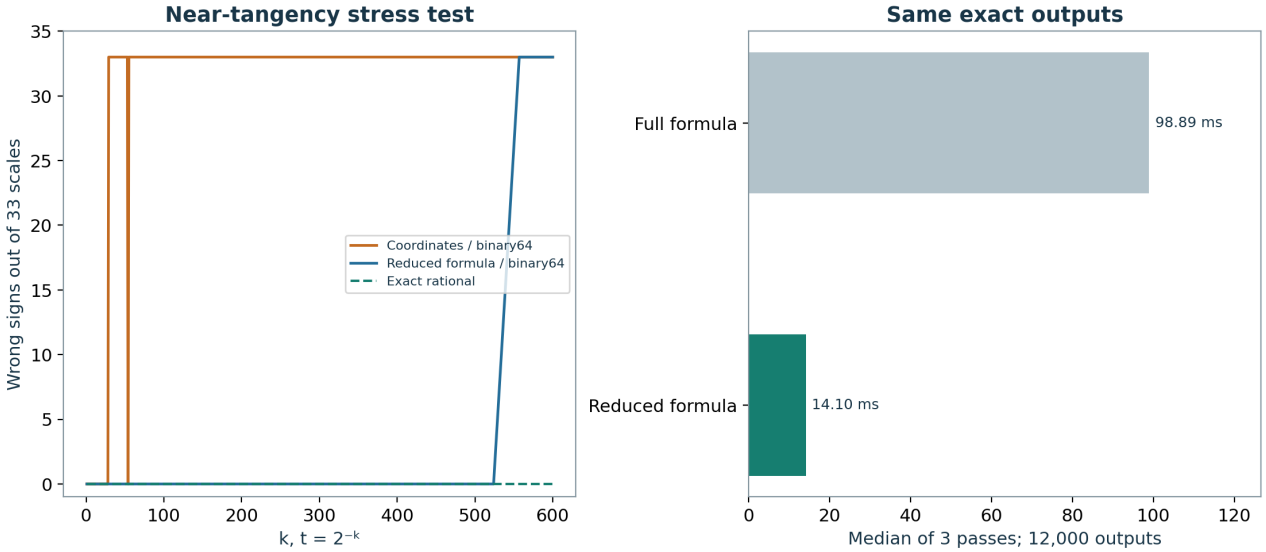


Figure 8. Left: incorrect signs by exponent k across 33 scales. Right: median times for two exact formulas returning the same output. The stress dataset was deliberately constructed near tangency.

The direct formula subtracts nearby values after rounding. The reduced formula avoids that subtraction and preserves the sign longer, until very small values round to zero. In the 1,200 radial controls, exact computation obtained every sign correctly, whereas binary64 disagreed in 1,095 cases. [E1]

These are not typical error rates for real systems. Every variant uses squares; none uses pi. The experiment therefore identifies effects of finite representation, conditioning and loss of depth. It does not attribute those effects to the existence of circles.

“Zero error” here means exact rational equality and correct sign within the tested model. Physical measurement precision and specification correctness are not automatically supplied by Fraction.

20. Efficiency from a proved identity

The generic kernel constructs T+t·d, computes its quadrance and subtracts ρ. The reduced kernel uses the proved identity Q(d)t². Both receive the same rational inputs and return the same quantity. Geometry precomputation is outside the timed region. [E1]

$$Q(T + td) - \rho = Q(T) - \rho + 2t(T \cdot d) + t^2 Q(d) = t^2 Q(d)$$

Pass	Generic exact (ms)	Reduced exact (ms)
1	99.361	15.326
2	98.892	14.000
3	98.109	14.104

Medians were 98.892 ms and 14.104 ms, a ratio of 7.012. All 12,000 outputs agreed exactly in three alternating-order passes. The initial run gave a ratio of 7.342. Timing variation is retained in the records; the output hash agrees across both runs.

The mechanism of improvement

Conditions Q(T)=ρ and T·d=0 eliminate repeated operations. The improvement follows from applying an identity under previously checked assumptions. Software can use the same mechanism: state invariants, verify that they hold, and reduce work within their valid domain.

The experiment measures only this kernel in this environment. It excludes file input, rendering, initial validation and model calls. The observed ratio is not a general productivity, token-use or AI-performance gain.

The cost of exact arithmetic

Fractions store numerators and denominators that can grow. Exactness removes rounding in these operations, but it does not remove memory cost, runtime or representation choices. An application may combine exact decisions with approximate visualization while preserving the boundary between them.

21. An executable bridge contract

The file codigo/nucleo_exato.py implements classification and proposal validation. Allowed inputs are integers or rational strings; floating-point values do not silently enter the kernel. The contract declares units and reference axis.

```
{
  "point": ["1", "3/2", "1/2"],
  "exit": ["0", "0", "7"],
  "rho": "7/2",
  "axis": ["0", "0", "1"],
  "csc2": "14",
  "units": {"point": "u", "exit": "u",
            "rho": "u^2", "axis": "1", "csc2": "1"}
}
```

Acceptance requires surface contact, zero tangency dot product, an exterior exit on the positive semiaxis, a nonzero direction and angular ratio 1/14. The certificate returns the exact supporting quantities. This domain is deliberately specific: a different geometry needs a different contract.

Output for the example	Certified value
Q(T) − ρ	0
T·(X − T)	0
Q(X − T)	91/2
Angular ratio s	1/14

Tests include displaced contact, incorrect dimension, an off-axis exit, altered radius, incompatible units, zero denominator, missing fields and out-of-schema actions. All 396 altered proposals were rejected by the expected condition. Formal contract validity does not establish the suitability of physical equipment.

The same principle can support an AI system: the model proposes, while a deterministic verifier decides only those properties it can certify. This architecture was illustrated with constructed inputs, without measuring model responses.

22. Low- and medium-complexity project prompts

Low complexity - exact geometric classifier

“Using one model, develop an offline Python tool that classifies a rational point as inside, touching or outside a sphere centered at the origin. Accept three coordinates and a positive squared radius as integers or fraction strings. Use exact fractions. Return the sign of $x^2+y^2+z^2-\rho$ and the rational value supporting the decision. Reject incorrect dimensions, zero denominators, floating-point input and incompatible units. Include tests on both sides of the surface and at contact. Deliver code, execution instructions and a result log. Scope ends at the classifier; add no networking, agents, rendering or extra features.”

Acceptance: compare each sign with an independent exact expression and preserve input and certificate. The verifiable objective is a correct decision within the formal domain without silent conversion.

Medium complexity - viewer that preserves depth

“Develop a standalone offline HTML/JavaScript file displaying tetrahedron, octahedron, sphere and bridge $T \to X$. Provide synchronized 3D and xy projection views. Include a button that moves to the view where the tetrahedron projects to a square with diagonals; allow locking the solid and independently changing the circle plane. Distinguish inscribed and circumscribed circles. Selecting an edge must display its spatial points, projection and the tangency condition actually computed. Keep exact values and certificates separate from drawing numbers. Identify intersections that exist only in the image. Deliver clear controls and tests of reference configurations. Use one model; access neither the internet nor external services.”

Acceptance: camera changes preserve certified spatial relations; actual object transformations recalculate them. The apparent diagonal crossing retains the distinction between $z=+c$ and $z=-c$. Circle centers, radii and planes are declared.

23. A complex project and quality measurement

Prompt - offline proposal and verification pipeline

“Build an offline trajectory-analysis system for explicitly specified geometries. Use one model throughout development. Separate inputs, units, geometric proposal, exact kernel, acceptance policy and visualization. Every simplification must state its enabling assumptions and return a checkable certificate. The kernel accepts rationals and the algebraic types actually implemented and tested; other input types return an unresolved result with a reason. Keep physical measurement uncertainty distinct from arithmetic error. Impose an explicit retry limit on automated correction and retain a complete decision log. Perform no external actions, access no network and add no out-of-scope functionality. Deliver specification, incremental implementation, representative and adversarial tests, stage timings and a limitations report. Each stage must satisfy criteria defined before implementation.”

How to test prompt improvements without assuming them

Dimension	Proposed measurement
Efficiency	Total time, verifier time, tokens if available, and attempts.
Quality	Requirements satisfied on cases withheld from method adjustment.
Scope	Extra features, omitted requirements and constraint violations.
Error	Semantic, arithmetic, integration and modeling failures, separately.
Rework	Corrections until acceptance or the retry limit.

A future evaluation should fix tasks, criteria and budget; compare a baseline instruction with the structured instruction; record the same model’s version and parameters; alternate order; repeat trials; and reserve unseen tasks for generalization. None of these model experiments is represented as having occurred in this edition.

A retry limit prevents an operational correction process from continuing indefinitely. This software control rule can be useful even when the underlying mathematical problem still lacks a termination proof within the study.

24. Exactness, philosophy and investigable applications

The proponent’s philosophical motivation values proportion, truth and examining a result closely. The square-on-the-diagonal example associated with the Meno offers an accessible picture of exact area doubling: the diagonal d of a square of side a satisfies $d^2=2a^2$. The discussion of image and likeness in the Sophist is used as motivation to distinguish appearance from structure. [D1, A1]

Here that motivation becomes operational requirements: distinguish the object and its projection, preserve data, state assumptions, prove identities and record what the implementation actually produced. Section 25 develops the author’s philosophical-mathematical interpretation: the proved passage can be chosen, and the model gives that choice an operational expression.

A practical research path

A next application can select a family of exact decisions, such as dot-product signs, incidence, orthogonality and comparisons of quadrances. Each should declare its domain, computational cost and certificate. Images help formulate and interpret the problem; certificates check selected properties.

The prototype supplies one example of this path: a spatial bridge, a collision predicate, an exact reduction, stress data and a contract. Application to AI models remains an engineering hypothesis to measure on specified tasks and models. This study did not test physical accuracy, equipment performance or equivalence between physical theories.

Three-dimensional geometry is suitable when the question depends on depth. Other problems may need two dimensions, more dimensions, time or variables other than positions. A model should preserve the information relevant to its application. Arithmetic exactness and model adequacy are complementary criteria.

The bridge’s applicability can be developed and assessed independently of a Collatz proof. This allows the geometric construction and computational results to be presented on their verifiable merits while the dynamical connection remains under investigation.

25. Philosophy, mathematics and compatible results

Theodorus was selected to investigate Collatz through an exact, extensible construction. Return was confirmed for every start through 10,000,000 and proved for the infinite family in Section 4; universal convergence remains an additional proof requirement. The spatial passage was proved within its own geometric model. These results address different properties and can coexist. [R13]

Matthew 19:23-26: the author's chosen reference

23 Then Jesus said to his disciples, "Truly I tell you, it is hard for someone who is rich to enter the kingdom of heaven.

24 Again I tell you, it is easier for a camel to go through the eye of a needle than for someone who is rich to enter the kingdom of God."

25 When the disciples heard this, they were greatly astonished and asked, "Who then can be saved?"

26 Jesus looked at them and said, "With man this is impossible, but with God all things are possible."

English wording supplied by the author; verse 26 emphasized. The initial verse label "3" is corrected to "23". No Bible edition was specified. [M19]

Depth and choice: the philosophical interpretation

The author draws a philosophical connection between the observed arithmetic return and the possibility of a spatial passage that can be chosen. In this reading, the words of Jesus express a relationship between possibility, knowledge and decision. The passage is not presented as a result that contradicts arrival at 1.

Invisibility has a defined geometric meaning: P(x,y,z)=(x,y) suppresses depth and places X=(0,0,7) and O at the same apparent center. The geometric and trigonometric demonstration certifies path B. Explicit choice between Γ_0=L and Γ_1=B, with c∈{0,1} and 0≤t≤1, has the following operational form:

$$P(\Gamma_c(t)) = (0,0) \quad \Longleftrightarrow \quad (c,t) = (1,1)$$

The equivalence follows from the two paths in Section 16. The model specifies path selection; it does not define a random process for evaluating access probabilities. Collatz remains the deterministic rule declared in the first part of the article.

The dialogue with Socrates and Jesus belongs to the author's philosophical interpretation. The Theodorus construction investigates the destination of integer trajectories; the spatial construction investigates existence and depth. Their compatibility follows from their definitions and the scope of each result.

26. Sources, reproduction and authorship statement

[D1] Davi Cançado Dutra Cota, Brazil. Hypotheses, direction and framework of this proposal: Theodorus as an extensible structure for testing Collatz; depth and a spatial passage as a complementary geometric study; philosophical interpretation of the results.

[A1] COTA. Proposed solution of the Collatz Conjecture through perfect squares and the mathematics of exactness. Supplied manuscript, 32 pp., 21 October 2026, declaring Claude assistance; geometry, pp. 6-12; Collatz, pp. 25-27.

[A2] Collatz, Teodoro e as portas quadradas: avaliação comparativa. Earlier 16-page document from this session; three enumerations through 1,000,000. Copy included in fontes/.

[T14] COTA. A espiral de Teodoro e o círculo de lado 1. Earlier supplied version, 14 pp., 21 October 2026; Theorem 12 and Proposition 15, p. 9; neighbouring rules, pp. 10-11. Unchanged copy in fontes/.

[E1] Original bridge protocol and execution: registros/protocolo_previo.json; codigo/testar_ponte.py; resultados/resumo.json. Includes 33,033 states, 19,800 stress cases, controls, contracts and runtimes.

[R11] codigo/verificar_positividade.py; resultados/positividade_v1_1/. Thirty thousand transitions in three representations, ten trajectories and 2,002 states of B and L.

[R12] codigo/teodoro_invariantes.py; resultados/autossimilaridade_v1_1/. Initial verification of 257 levels and the recurrence/square figure; expanded in [R13].

[R13] codigo/ampliar_estudo.py, collatz_expansao.c and verificar_certificado_expansao.py; resultados/expansao_v1_1/. Ten million starts in three methods, a separate verifier, the infinite family and extreme scales. Complete protocols and data are included.

[R14] codigo/comparar_arquimedes.py and figura_arquimedes.py; resultados/arquimedes_v1_1/. One million transitions, one thousand trajectories, twenty thousand geometric pairs and exact rational bounds. Local comparison without external research.

[M19] Matthew 19:23-26. Portuguese and English wording supplied by the author, preserved in fontes/mateus_19_23_26_fornecido.txt and fontes/matthew_19_23_26_supplied_en.txt. Verse 26 is bold; the initial English verse label is corrected from 3 to 23. No Bible editions were specified.

Reproduction: LEIA_ME.txt contains dependencies, commands and formats. Reruns preserve canonical records. Code, data and received sources accompany the PDFs.

Contributions: Davi Cançado Dutra Cota structured the hypotheses, directed the project, and formulated and developed the methodological and theoretical-mathematical framework of this proposal. ChatGPT performed computational execution, test implementation and execution, verification of records and writing of this version under the author’s direction, using one model, no agents and no internet research.